F26 AIRPORT INNOVATION CHALLENGE · TRACK 01

AutoTrack

A solution to the baggage problem

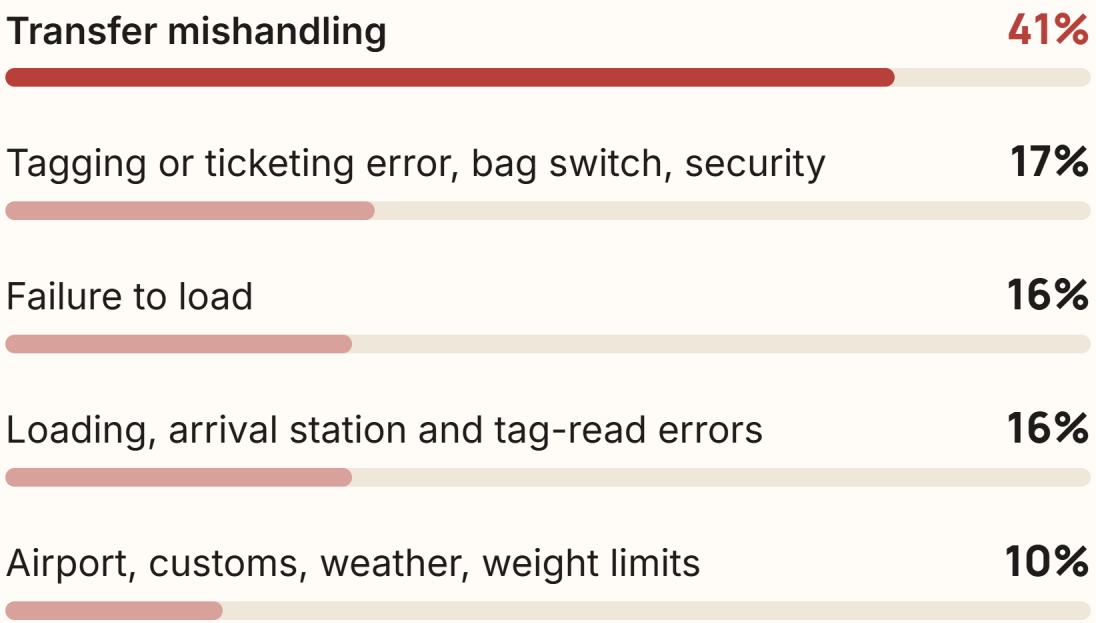
Alex · Joseph · Daniel · Ocean · Prem

Daniel Solutions

THE PROBLEM

Mishandled Luggage

WHAT CAUSES IT · SHARE OF MISHANDLED BAGS



116k
mishandled bags a year at YYZ (est.)

$367
CAD per bag, $896 if lost

$42.5M
CAD a year at YYZ (est.)

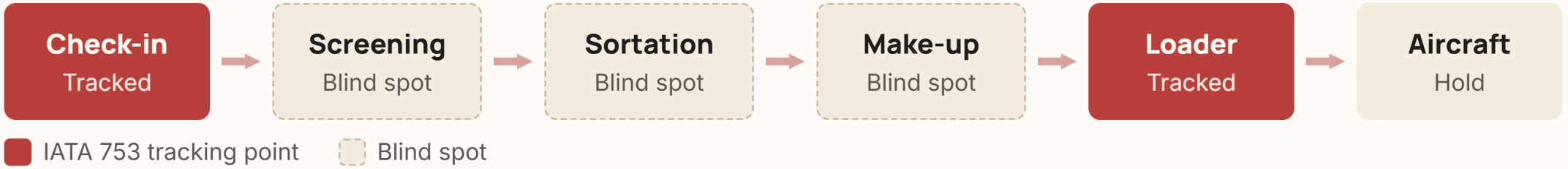
SITA 2025 & 2026 · GTAA 2025 · BoC 1.4113 · YYZ est. = ½ × 47.3M pax × 4.9/1,000 × $367 CAD

THE PROBLEM

Where bags go dark

IATA Resolution 753 tracks each bag at four points. Between them, nobody knows where it is.

A DEPARTING BAG'S JOURNEY AT YYZ



01

Where is the bag right now?

02

Did it reach the right plane?

03

Is the tag still on the right bag?

THE SOLUTION

How AutoTrack works

MESSAGES IN

DCS	bag + flight (BSM)
BLE · UWB	tag position
RFID	loader portal reads
BHS	jams · scale · photo
AODB	gate · close time
Loader pad	tag released

AUTOTRACK ENGINE

Zone + position

Right flight?

RFID ↔ tag match

ETA vs cut-off

SecureBag photo

Every read → IATA 753 event

ALERTS OUT

WRONG AIRCRAFT	light + beep
MISROUTED	pull bag
IDENTITY FLAG	human review
NOT SEEN / LATE	expedite
TAG NOT RETURNED	before close
ON TRACK	passenger app

→ Ops · handheld · tag · SmartBag

TAG LOOP · NEVER LEAVES YYZ

Check-in — Clip on → **Bag hall** — BLE zones → **Make-up** — UWB burst → **Loader pad** — Right flight? → **Return bin** — Tag back → **Reissue** — Next bag

FEASIBILITY

Built to fail safe

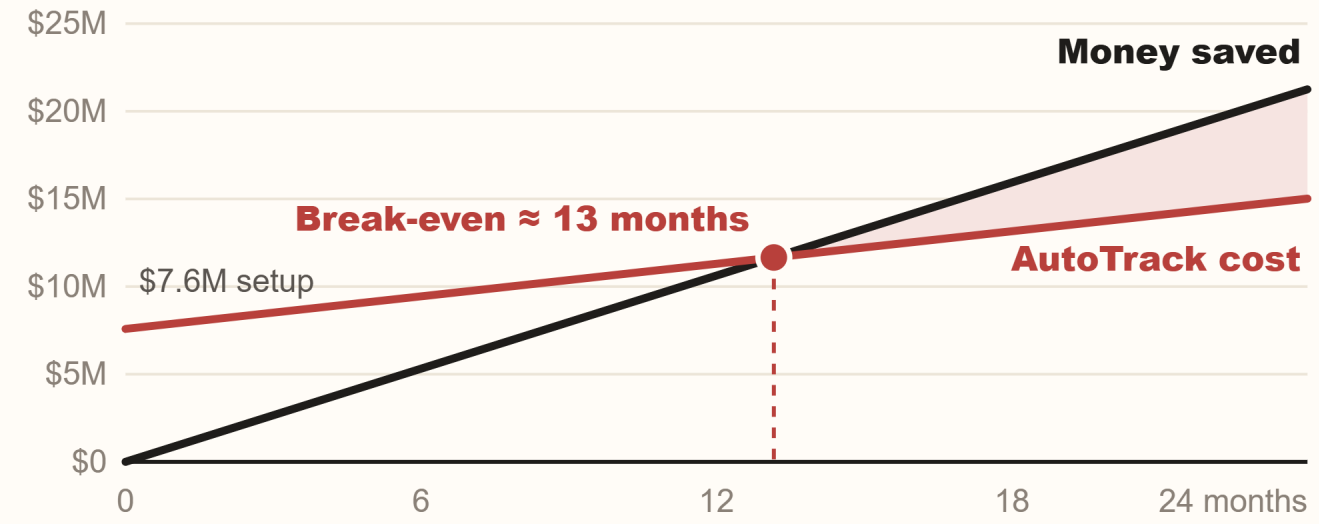
WHAT GOES WRONG	WHAT AUTOTRACK DOES	RECOVERY
Tag falls off	Paper tag still read at the loader	→ **Photo match reunites bag and tag**
Receiver down	Nearby receivers still hear the tag	→ **Shows “zone offline”, not “bag lost”**
Network down	Loader still reads the paper tag	→ **Falls back to today’s scanning**
Tag snags	Flat, rounded disc, no loose straps	→ **Belt jam alarms work as today**
Tag left on	Bag reaches the loader with tag on	→ **Alert before the hold door closes**

If any part fails, YYZ falls back to today’s process: never worse than now.

FEASIBILITY

Costs ~$4–5M a year. Saves far more.

COST VS SAVINGS OVER TIME · CAD, EST.



25% fewer mishandled bags · US$52 tag shown (US$32 tag: ~10 months)

~1 year

to pay for itself if mishandled bags drop 25% (10–13 months)

$6.2–7.6M

one-time setup

$2.9–3.7M

a year to run

Team estimates in CAD at BoC 1.4113 · AutoTrack = running cost + setup over 5 years · SITA 2026 · GTAA 2025

FEASIBILITY

Safe on the ground and in the air

Worker safety	Never controls the belts · smooth tag, nothing to snag	Canada Labour Code Part II
Battery	Tag comes off before the plane is loaded · small coin cell, within airline limits	IATA Dangerous Goods Regulations
Radio	Approved for use in Canada · low power, short bursts	ISED RSS-220
Security	Tag stays outside the bag · security screening works as it does today	Aviation Security Regs, 2012
Privacy	No passenger name on the tag · data deleted once the bag is delivered	PIPEDA
Human review	Staff check every alert · no bag is rejected automatically	IATA Resolution 753

IATA DGR · Panasonic CR2477 data sheet · ISED · Qorvo 2025 · PIPEDA · Canada Labour Code

RESULTS

A real step up, and it works

> **AutoTrack builds on what Pearson already runs: RFID reads, the baggage system and today's processes.**
>
> Our working prototype tracks a bag zone by zone and flags the wrong plane before takeoff.

Bags will always go missing sometimes. AutoTrack catches it before the plane leaves.

APPENDIX

Cost breakdown

SETUP (ONE TIME)

Item	Basis	CAD
Tags	40,000 × US$32–52	$1.81–2.96M
UWB anchors	240 × US$579	$0.20M
Loader stations	104 stations	$0.40M
BLE receivers	300 × US$78	$0.03M
Installation	= hardware cost	$0.63M
Software + integration	assumed	$2.10M
Contingency	20%	$1.03–1.26M
Total		$6.2–7.6M

RUNNING (PER YEAR)

Item	Basis	CAD
Lost tags	23,650 tags (0.1%)	$1.07–1.75M
Battery retirements	~4,800 tags/yr	$0.22–0.36M
Handler time	8 s/bag at $24/h	$1.26M
Maintenance	10% of hardware	$0.06M
Software ops	US$200k	$0.28M
Total		$2.9–3.7M

Team estimates · CAD at BoC 1.4113 · 23.65M bags/yr = ½ × 47.3M passengers × 1 bag

APPENDIX

Sources and assumptions

FACTS AND FIGURES

SITA Baggage IT Insights 2025 & 2026: 4.9 per 1,000 passengers, US$260 per bag, cause split
GTAA 2025 annual results: 47.3M passengers
Bank of Canada: 1.4113 CAD per US$ (Nov 6, 2026)
IATA Resolution 753: bag tracking points
IATA dangerous goods: 0.3 g lithium limit
Panasonic CR2477 SDS: 0.29 g lithium
Qorvo (Aug 2025): DWM3001C certified by ISED
ISED RSS-220 · PIPEDA · Canada Labour Code Part II · Aviation Security Regs 2012
Job Bank: Toronto ramp attendant $24/h

UNIT PRICES AND ASSUMPTIONS

Tag BOM US$32.00–52.48 (Qorvo DWM3001C module)
UWB anchor US$579 (UbiTrack A1, RobotShop)
RFID portal US$2,270 (Impinj R700, Zebra, Axis)
BLE receiver US$78 (Minew G1)
Release pad ~US$300, return bin ~US$150 (est.)
½ of passengers depart, 1 checked bag each
40,000 tags: peak hour × 5 h per trip × 1.2 spare
0.1% of tags fly (TAG NOT RETURNED alert)
Battery ~3 years · 25% fewer mishandled bags for payback

AI USE

Claude (Anthropic) helped with research, cost calculations and slide drafting; the team checked every figure. Full disclosure: aidframework.org [add link]